

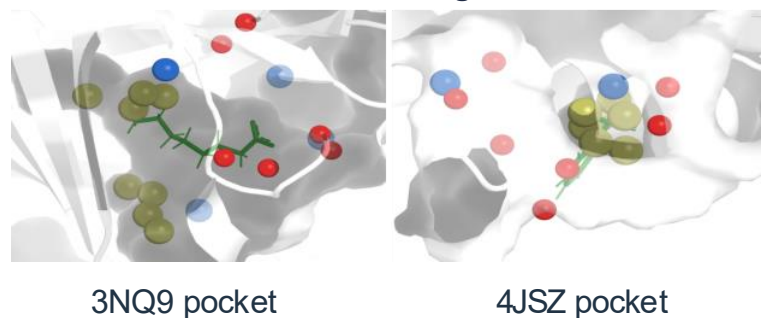
INPUT & FORMULATION

(Stages 1-2)

Protein-Ligand Input

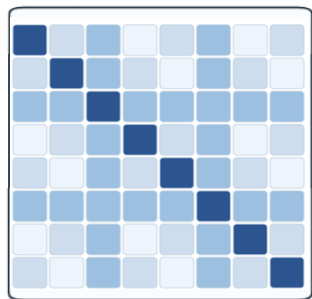


Discretized Binding Pocket



FAM-QUBO Formulation

$$E(x) = E_{score}(x) + K_{mono}E_{mono}(x) + K_{dist}E_{dist}(x)$$



K_{dist} penalty

distance consistency

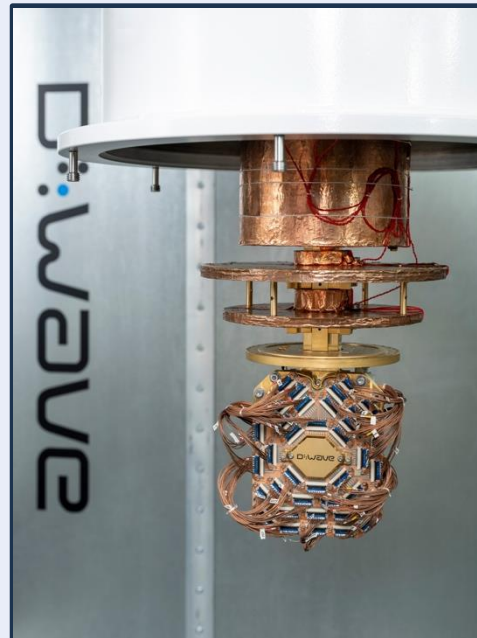
K_{mono} penalty

monogamy (1-to-1)

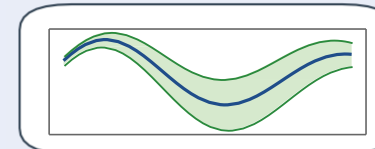
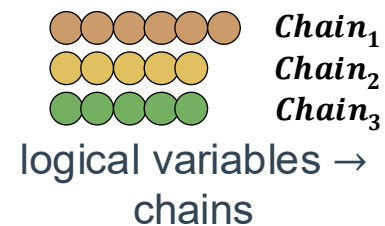
OPTIMIZATION LOOP

(Stages 2-3)

D-Wave Advantage2



Zephyr Topology
(minor-embedding)



GP posterior
(mean / 95% CI)

Expected Improvement
Acquisition
select next
(K_{dist} , K_{mono})

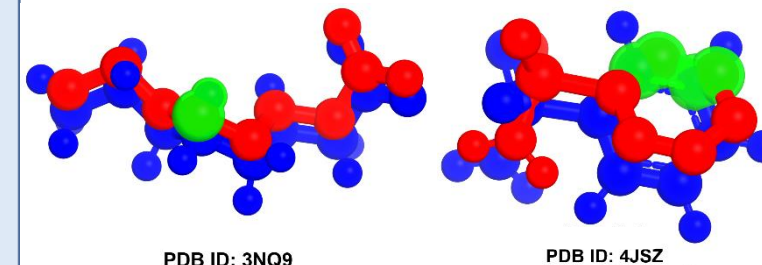
Penalty Parameter
Optimization
(Bayesian
Optimization)

Hardware
Tuning Sweep
(controlled-variable
design)

EVALUATION

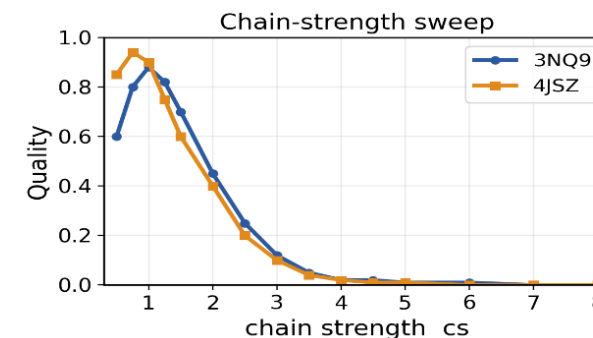
(Stage 4)

Decoded Poses



Native(crystal) QPU prediction Overlapped

Performance Metrics



Feasibility-gated
reporting

Validity Quality $TTS_{0.99}$ mRMSD